

Cell Theory

Cell Structure & Function

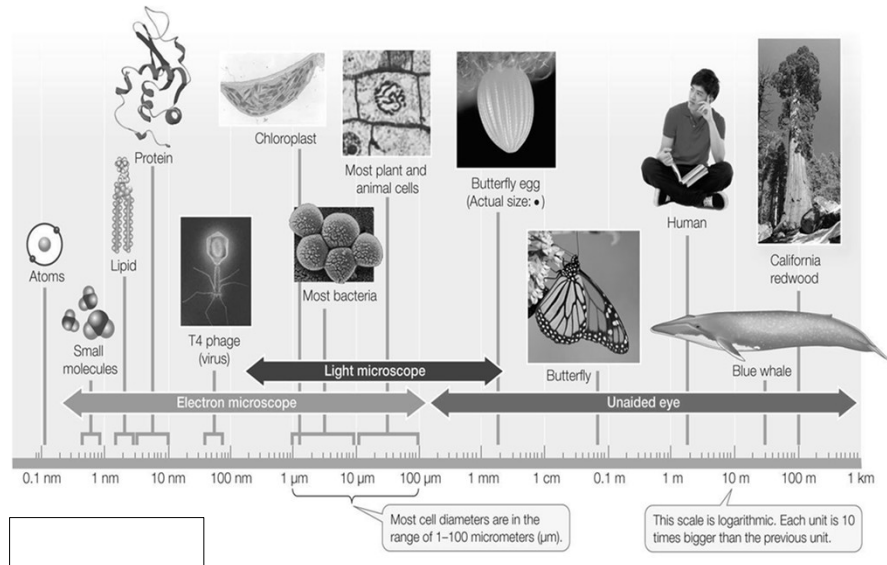
Eukaryotic Cells



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Cell size

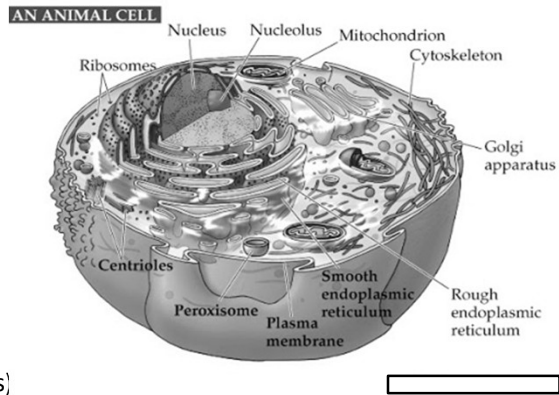


Purves *et al.* (1998). Life the Science of Biology, Sinauer Associates, Inc.

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Animal Cell

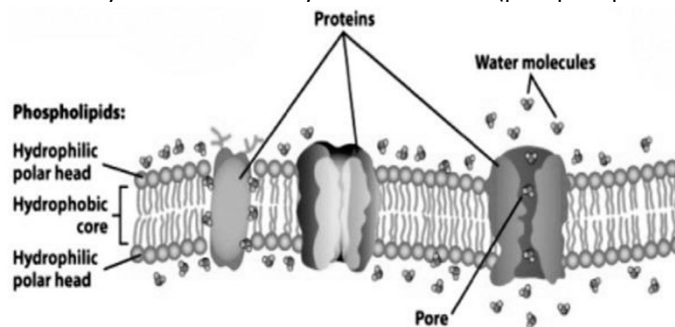
- Cell membrane
- Cytoplasm
- Mitochondria
- Endoplasmic Reticulum (ER)
- Golgi Apparatus
- Ribosomes
- Nucleus
- Nucleolus
- Chromatin
- Centrosomes (Centrioles)
- Lysosomes
- Microvilli (sing. Microvillus)
- Peroxisomes (or microbodies)



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Cell Membrane

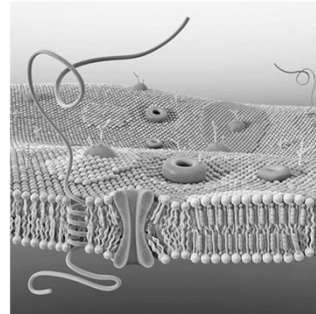
- ❖ Plasma/cell/ semipermeable/ selectively permeable membrane. It is outermost layer and is a double layer of lipids and proteins that surrounds a cell and separates the cytoplasm. It can also control the amount of some substances that go into or out of the cell.
- ❖ Phospholipids are a main component of the cell membrane. These are lipid molecules made up of a phosphate group head and two fatty acid tails. The properties of phospholipid molecules allow them to spontaneously form a double-layered membrane (phospholipid bilayer).



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Cell Membrane

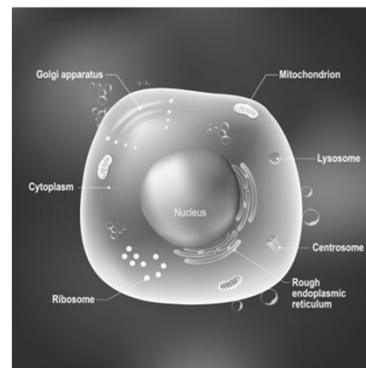
- ❖ Cell membrane has **transmembrane proteins**, which are embedded in the membrane but stick out on both sides. Some of these proteins are receptors which bind to signal molecules, while others are **ion channels** which are the only means of allowing ions into or out of the cell.
- ❖ In addition, the cell membrane contains **glycolipids and sterols**. One important sterol is **cholesterol**, which regulates the fluidity of the cell membrane in animal cells. When there is less cholesterol, membranes become more fluid, but also more permeable to molecules. The amount of cholesterol in the membrane helps maintain its permeability so that the right amount of molecules can enter the cell at a time, not too many or too few.



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Cytoplasm

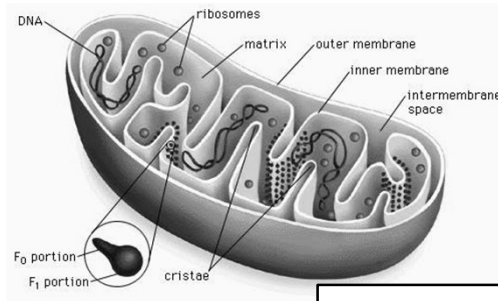
- ❖ Cytoplasm refers to the fluid that fills the cell, which includes the **cytosol** along with **filaments, proteins, ions and macromolecular structures** as well as the organelles suspended in the cytosol. **It holds the organelles in place within the cell.**
- ❖ Site of many biochemical processes - **glycolysis, protein synthesis.**
- ❖ Cytoplasm is a complex and crowded system containing a wide range of particles – from ions and small molecules, to proteins to giant multi protein complexes and organelles.



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Mitochondria

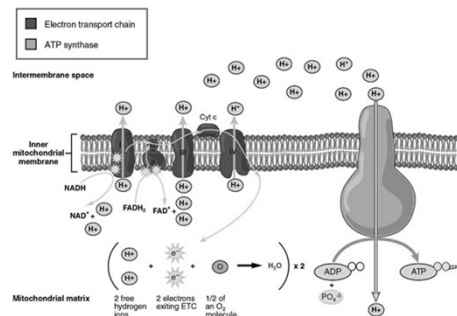
- ❖ They are a double membrane structure. The outer membrane is smooth, but the inner membrane is folded inwardly to form extended projections called cristae.
- ❖ The two membranes produce two inner compartments, the inter-membrane space between the outer and inner membranes and the region enclosed by the inner membrane, called the matrix.
- ❖ Some of the steps of cellular respiration occur in the matrix of the mitochondrion. Thus, there are many enzymes found in the matrix as well as on the inner membrane.



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Mitochondria...

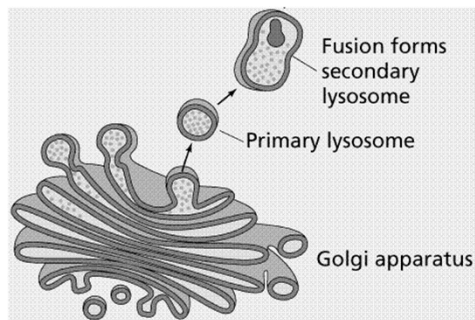
- ❖ The process of cellular respiration occurs in the mitochondria. During this process, sugars and fats are broken down and energy is released in the form of adenosine triphosphate (ATP). ATP powers all cellular processes, and mitochondria produce a cell's ATP, so mitochondria are commonly known as "the **powerhouse** of the cell".
- ❖ The aerobic steps of respiration occur here – **Krebs Cycle and Electron Transport System**.
- ❖ The quantity of mitochondria within cells varies, the more energy a cell needs, the more mitochondria it contains.



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Golgi Apparatus

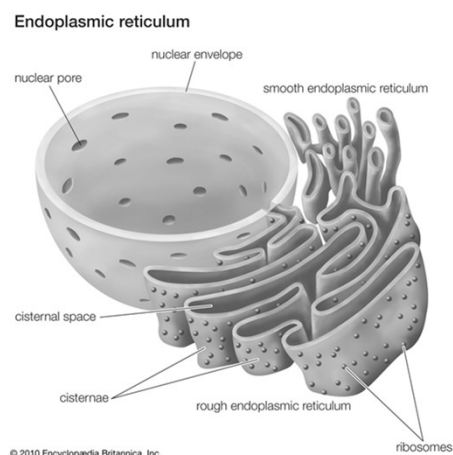
- ❖ Called the "**post office**" of the cell or "**packaging organelle**" because it plays a role in transporting proteins.
- ❖ It is usually connected to an endoplasmic reticulum (ER) because it stores and then transports the proteins produced in the ER



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Endoplasmic Reticulum (ER)

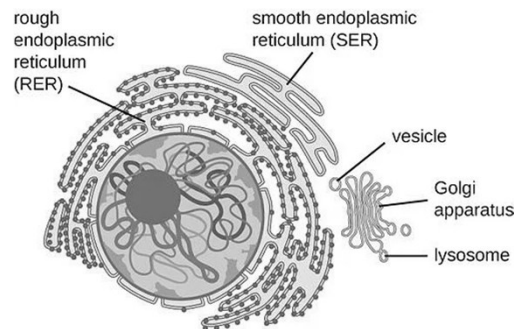
- ❖ Single membrane structure. The membrane of the ER is organized into a network of membranous tubules and sacs called cisternae.
- ❖ The endoplasmic reticulum is where **proteins and lipids are produced** within the cell, and is also concerned with the transport of these materials within the cell.
- ❖ The smooth and rough regions of the ER are interconnected.



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Endoplasmic Reticulum (ER)

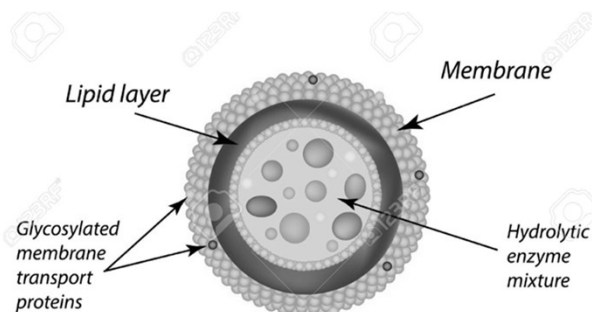
- ❖ R-ER - "Rough" indicates that there are ribosomes attached to the surfaces of the endoplasmic reticulum. The RER is in fact an extension of the outer nuclear membrane. Functions of the RER include anchorage of newly synthesized proteins, and the finishing of proteins – Smooth ER (SER): "Smooth" indicates that there are no ribosomes attached to the surfaces of the endoplasmic reticulum. It is a network of pipe-like interconnected tubes. Functions of the SER include synthesis of lipids, processing of sugars, and detoxification of drugs and poisons



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Lysosomes

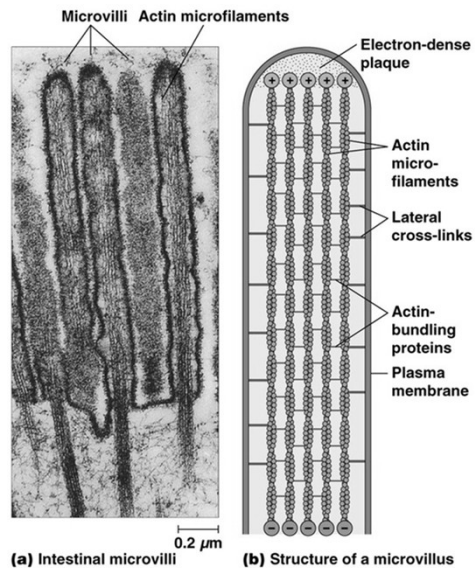
- ❖ Lysosomes are tiny sacs filled with digestive or hydrolytic enzymes that enable the cell to utilize its nutrients.
- ❖ These hydrolases break down nucleotides, proteins, lipids, phospholipids, and also remove carbohydrate, sulfate, or phosphate groups from molecules.
- ❖ Responsible for 'cell death' - burst to carry out 'autolysis'.
- ❖ Single membrane structure



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Microvilli (sing. Microvillus)

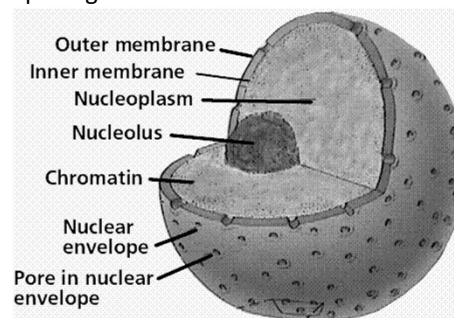
- ❖ Microvilli are finger-like projections on the outer-surface of the cell.
- ❖ Not all cells have microvilli. Their function is to increase the surface area of the cell, which is the area through which diffusion of materials both into, and out of, the cell is possible.



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Nucleus

- Contains DNA (genetic information), which is organized into chromatin. When the cell prepares to divide, the chromatin begins to condense into recognizable structures called chromosomes.
- One of the main functions of the nucleus is to control the production of proteins in the cell.
- **Nuclear Membrane** – The nuclear membrane separates the nucleus and the nucleolus from the rest of the contents of the cell. It's a double membrane structure. In this membrane, there are holes, called nuclear pores, which regulate the passage of materials in and out of the nucleus.



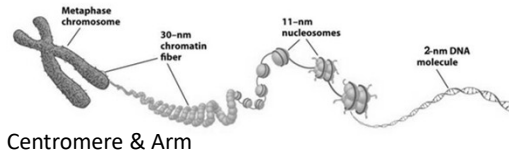
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Nucleus...

- **Chromatin** is very fine thread-like combinations of DNA and protein in non-dividing nuclei. The protein assists in the efficient packaging and regulation of DNA activity. A human nucleus contains 46 such fine threads of chromatin.
- **Nucleolus** – A region of densely stained fibers and granules adjoining chromatin. The nucleolus is responsible for the synthesis of cell organelles (e.g. lysosomes, ribosomes, etc.). – In the nucleolus, rRNA is synthesized and assembled with proteins from the cytoplasm to form ribosomal subunits. The subunits pass from the nuclear pores to the cytoplasm where they combine to form ribosomes.



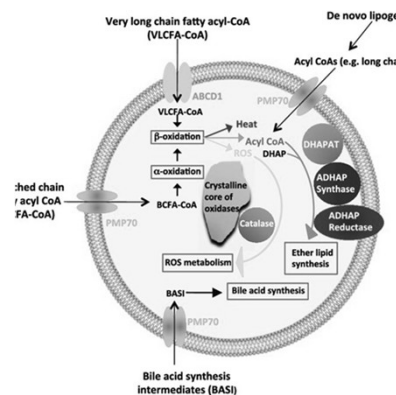
Levels of DNA Packaging



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Peroxisomes or microbodies

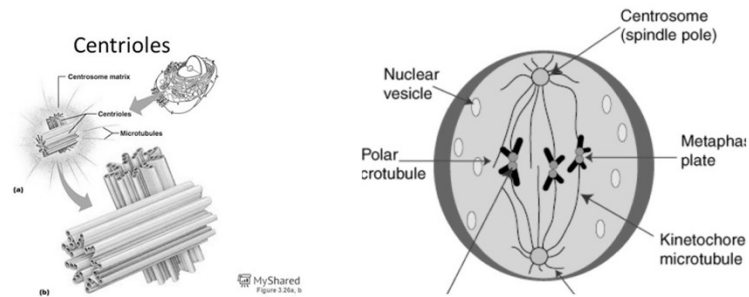
- ❖ Membrane-bound vesicles containing oxidative enzymes that generate and destroy hydrogen peroxide. They are involved in the catabolism of very long chain fatty acids, branched chain fatty acids, D-amino acids and polyamines.
- ❖ Other functions - glyoxylate cycle in germinating seeds ("glyoxysomes"), photorespiration in leaves.



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Centrosomes (centrioles)

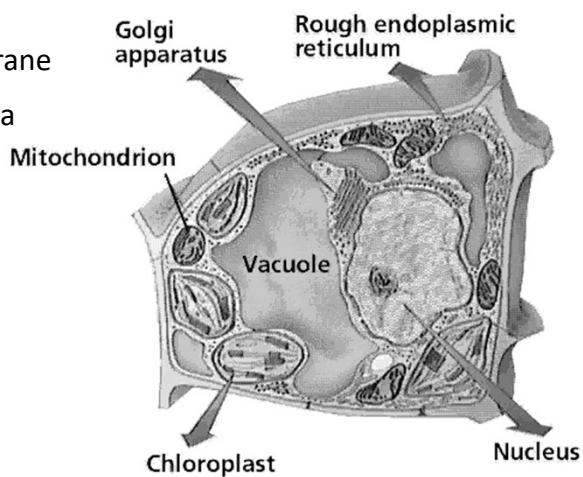
- The centrosomes contain the centrioles, which are responsible for cell-division. Centrioles divide to form spindle fibers during cell division.
- Centrioles are cylindrical structures that are composed of groupings of microtubules arranged in a 9 + 3 pattern. The pattern is so named because a ring of nine microtubule "triplets" are arranged at right angles to one another.
- Centrioles replicate during the interphase stage of mitosis and meiosis.



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Plant Cell – Structure & Function

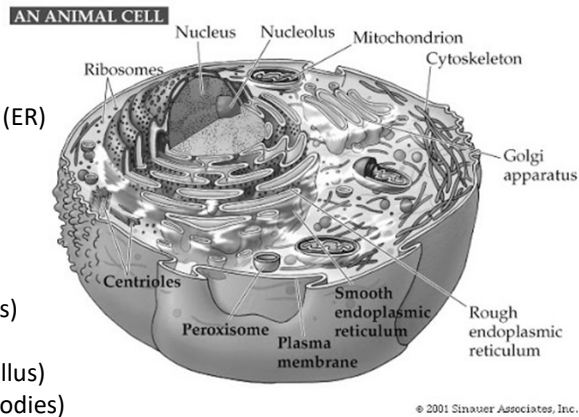
- Cell Wall
- Plasma membrane
- Plasmodesmata
- Vacuole
- Microtubules
- Chloroplasts



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Animal Cell

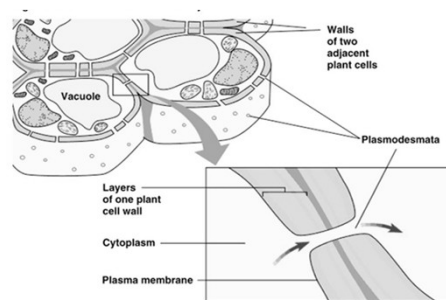
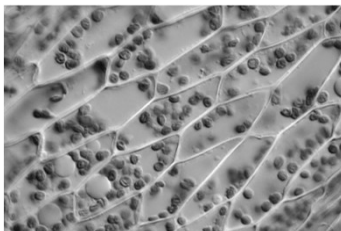
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Cell wall

- Plant cells have a thick (4-20 μ m) cell wall which composed of cellulose, Hemi-cellulose, pectin and lignin molecules.
- It provides strength to the plant. Visible under light microscope.



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Cell Wall

Structure - Cell Wall is a thick, rigid structure of around 4-20um; Cell wall does not have receptors. – it is present in plant cell, bacterial, algae and fungal cell, helps in providing shape and rigidity to the cell. – Metabolically Inactive and non-living.

Composition - Made up of cellulose (plant cell), chitin (fungi), and peptidoglycan (bacteria).

Functions - Cell wall provides rigidity to the cell, protect the cell from external shock and other mechanical forces. – It is responsible for cell morphology. – It prevents large molecule to enter into the cell.

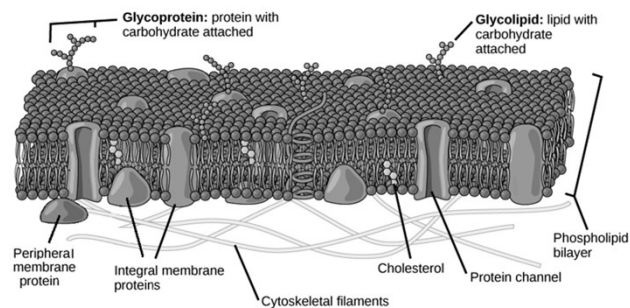
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Plasma membrane (plasmalemma)

Structure: It is a delicate, thin structure of around 5-10nm wide and is visible under electron microscope, and it is present in all types of cell, helps in the passage of molecules and protects the protoplasm.

Composition: Made up of lipids and proteins and carbohydrates.

Functions: – The plasma membrane is flexible enough to move closer to or away from the cell wall - according to changes in the water content of the cytoplasm within the cell. – As a differentially-permeable surface, the plasma membrane controls movement of solutes in and out of the cell.



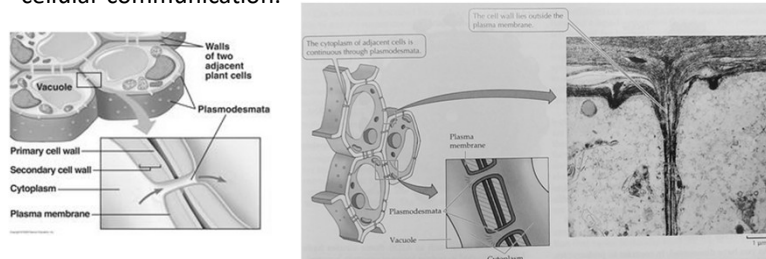
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Plasmodesmata

Structure: Plasmodesmata (singular, plasmodesma) are tiny strands of cytoplasm that pass through pores in plant cell walls, forming "connections" or "pathways" between adjacent cells.

A majority of plasmodesmata also contain a narrow tube-like structure called the **desmotubule**, which is derived from the **SER** of the connected cells.

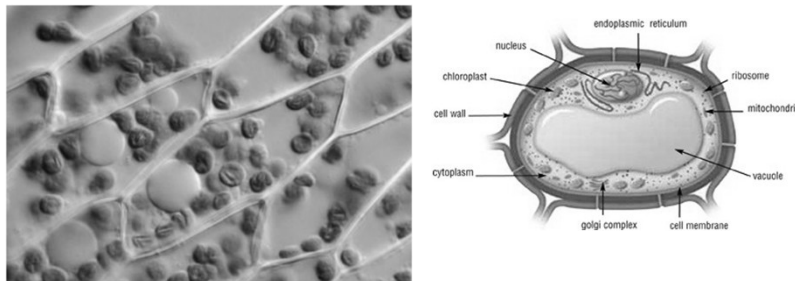
Function: – These cell-cell connections are especially important for the survival of plant cells during conditions of drought. – They allow certain molecules to pass directly from one cell to another and are important in cellular communication.



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Vacuoles

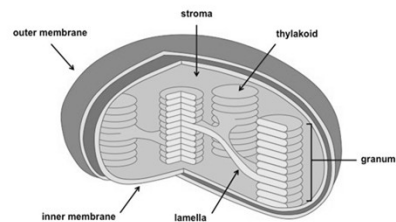
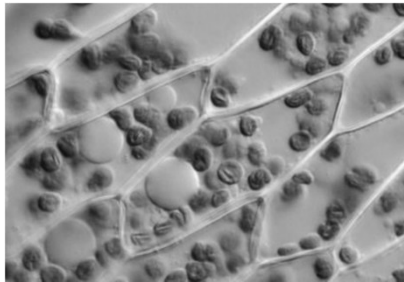
- ❖ A cell's vacuole can occupy a large proportion of the total volume of the cell - e.g. 90% of the volume of some mature plant cells.
- ❖ Each vacuole is enclosed by a vacuolar membrane called the tonoplast.
- ❖ Contents of the vacuole: – Cell sap, which is a solution of salts, sugars and organic acids. – Enzymes needed for recycling components of cells, e.g. chloroplasts. – **Anthocyanins** are sometimes present in cell vacuoles. These are chemical pigments responsible for some of the (non-green) colours of flowers, e.g. reds, blues, purples.



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Chloroplast

- ❖ Chloroplasts are the sites of photosynthesis and double membrane structure.
- ❖ These two membranes create a thin inter-membrane space between the membranes, and a larger space surrounded by the inner membrane.
- ❖ Within the larger space is another membrane system arranged into **flattened sacs called thylakoids**. The thylakoids are surrounded by a fluid called the **stroma**.



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Chloroplast

- ❖ The thylakoid membranes contain chlorophyll and other photosynthetic pigments arranged in photosystems, together with stalked particles, and is the site of photosynthesis and ATP synthesis
- ❖ Chloroplasts are very important parts of plant cells. Some cells include up to 50 chloroplasts.
- ❖ The number of chloroplasts per cell varies according to the type of cell and its function. They are plentiful in leaf cells that receive sunlight - as opposed to root cells that do not receive light.

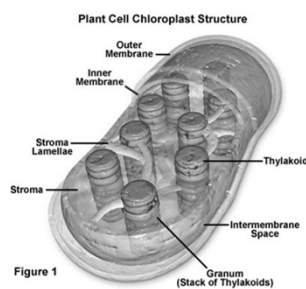
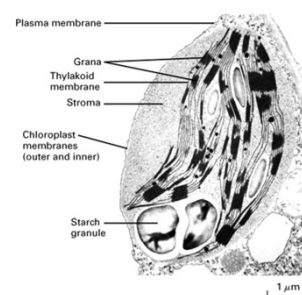


Figure 1



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Property	Prokaryotic Cells	Eukaryotic Cells
Size	Small Cells (< 5µm)	Larger cells (> 10 µm)
Cell type	Always unicellular	Often multicellular
Nucleus	Absence of organized nucleus, nuclear membrane, nucleolus, nucleoplasm and chromatin fibers	Presence of organized nucleus, nuclear membrane, nucleolus, nucleoplasm and chromatin fibers
Organelles	No membrane bound organelles present like ER, Golgi complex, mitochondria etc.	Membrane bound organelles presents such as ER, Golgi bodies, mitochondria etc.
DNA	Usually is circular, without histone	DNA is linear and associated with histone to form chromatin
Ribosomes	Small (70S type - 50S & 30S)	Large (80S type - 60S & 40S)
Cytoskeleton	No Cytoskeleton	Always has a cytoskeleton
Motility	Motility by rigid rotating flagellum (made up of flagillin)	Motility by flexible waving cilia or flagella (made up of tubulin)
Cell division	Cell division is by binary fission	Cell division by mitotic or meiosis
Reproduction	Reproduction is always asexual	Reproduction is sexual or asexual
Vacuoles	Gas filled vacuoles may present	Sap filled vacuoles
Respiratory enzymes	Located on cell/plasma membrane and mesosome	Located in mitochondria and cytoplasm
Examples	Bacteria, mycoplasma, cyanobacteria	All animal and plant cells

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Property	Plant Cells	Animal Cells
Size	Usually larger than animal cells	Usually smaller than plant cells
Cell wall	In addition to plasma membrane CW is present and consist of middle lamellae, primary and secondary walls	Absent
Nucleus	Present along the periphery of the cell	Present at the centre of the cell
Plasmodesmata	Present	Absent
Chloroplast	Present	Absent
Vacuole	Large and permanent	Small and temporary
Storage	Storage material is starch granules	Glycogen granules

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THANK YOU